kavyavaish0

October 29, 2025

[1]:
```
pip install pandas
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in
c:\users\hp\appdata\roaming\python\python313\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas) (2.3.4)
Requirement already satisfied: python-dateutil>=2.8.2 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas)
(2025.2)
Requirement already satisfied: six>=1.5 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

Pandas

[2]:
```
import pandas as pd
```

[3]:
```
df = pd.read_csv(r"c:\Users\hp\Documents\indian_sales_performance_dataset.csv")
```

[4]:
```
pd
```

[4]:
```
<module 'pandas' from 'C:\\Users\\hp\\AppData\\Roaming\\Python\\Python313\\site-
packages\\pandas\\__init__.py'>
```

[5]:
```
df
```

[5]:
```
   Employee_ID Employee_Name  Gender  Age Department         Region  \
0         3001        Aarohi  Female   36      Sales          Assam
1         3002        Ananya  Female   35         IT          Bihar
2         3003        Ishita  Female   45    Finance  Uttar Pradesh
```

```
3            3004         Vihaan     Male    56  Marketing  Uttar Pradesh
4            3005          Dhruv     Male    31    Finance         Punjab
..            ...            ...    ...   ...        ...            ...
195          3196         Aniket     Male    26    Finance    Uttarakhand
196          3197          Dhruv     Male    23         HR         Odisha
197          3198          Aryan     Male    36    Finance          Assam
198          3199        Karthik     Male    27    Finance          Assam
199          3200          Sneha   Female    34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                ...            ...                    ...                ...
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[6]:
```
df.head()
```

[6]:
```
   Employee_ID Employee_Name  Gender  Age Department         Region  \
0         3001        Aarohi  Female   36      Sales          Assam
1         3002        Ananya  Female   35         IT          Bihar
2         3003        Ishita  Female   45    Finance  Uttar Pradesh
3         3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4         3005         Dhruv    Male   31    Finance         Punjab

   Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                 2          48648                      2            Average
1                13          75729                      3               Good
2                25          66971                      1            Average
3                18          43658                      6               Good
4                 4          46253                      7          Excellent
```

[7]:
```
df.tail()
```

[7]:
```
     Employee_ID Employee_Name  Gender  Age Department       Region  \
195         3196        Aniket    Male   26    Finance  Uttarakhand
196         3197         Dhruv    Male   23         HR       Odisha
197         3198         Aryan    Male   36    Finance        Assam
198         3199       Karthik    Male   27    Finance        Assam
```

```
199         3200         Sneha  Female   34         HR  West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent
```

[8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 10 columns):
 #   Column                 Non-Null Count  Dtype 
---  ------                 --------------  ----- 
 0   Employee_ID            200 non-null    int64 
 1   Employee_Name          200 non-null    object
 2   Gender                 200 non-null    object
 3   Age                    200 non-null    int64 
 4   Department             200 non-null    object
 5   Region                 200 non-null    object
 6   Experience_Years       200 non-null    int64 
 7   Monthly_Sales          200 non-null    int64 
 8   Customer_Satisfaction  200 non-null    int64 
 9   Performance_Rating     200 non-null    object
dtypes: int64(5), object(5)
memory usage: 15.8+ KB
```

[9]:
```
df.describe()
```

[9]:
```
       Employee_ID        Age  Experience_Years  Monthly_Sales  \
count   200.000000  200.00000        200.000000     200.000000   
mean   3100.500000   38.59500         14.920000   81435.635000   
std      57.879185   10.82529          8.673508   35862.006242   
min    3001.000000   21.00000          0.000000   25155.000000   
25%    3050.750000   29.00000          7.000000   51016.500000   
50%    3100.500000   38.00000         16.000000   80172.000000   
75%    3150.250000   48.00000         23.000000  110201.250000   
max    3200.000000   57.00000         29.000000  149661.000000   

       Customer_Satisfaction  
count             200.000000  
mean                5.370000  
std                 2.883518  
min                 1.000000  
25%                 3.000000  
```

```
50%                     5.000000
75%                     8.000000
max                    10.000000
```

[10]:
```
df.shape
```

[10]:
```
(200, 10)
```

[11]:
```
df.columns
```

[11]:
```
Index(['Employee_ID', 'Employee_Name', 'Gender', 'Age', 'Department', 'Region',
       'Experience_Years', 'Monthly_Sales', 'Customer_Satisfaction',
       'Performance_Rating'],
      dtype='object')
```

[12]:
```
df.values
```

[12]:
```
array([[3001, 'Aarohi', 'Female', …, 48648, 2, 'Average'],
       [3002, 'Ananya', 'Female', …, 75729, 3, 'Good'],
       [3003, 'Ishita', 'Female', …, 66971, 1, 'Average'],
       …,
       [3198, 'Aryan', 'Male', …, 75939, 10, 'Good'],
       [3199, 'Karthik', 'Male', …, 54516, 2, 'Excellent'],
       [3200, 'Sneha', 'Female', …, 81526, 3, 'Excellent']],
      shape=(200, 10), dtype=object)
```

[13]:
```
df.dtypes
```

[13]:
```
Employee_ID               int64
Employee_Name            object
Gender                   object
Age                       int64
Department               object
Region                   object
Experience_Years          int64
Monthly_Sales             int64
Customer_Satisfaction     int64
Performance_Rating       object
dtype: object
```

[14]:
```
df.loc[0]
```

[14]:
```
Employee_ID                 3001
Employee_Name             Aarohi
Gender                    Female
Age                           36
Department                 Sales
```

```
Region                        Assam
Experience_Years                  2
Monthly_Sales                 48648
Customer_Satisfaction             2
Performance_Rating          Average
Name: 0, dtype: object
```

[15]:
```
df.iloc[0,1]
```

[15]:
```
'Aarohi'
```

[16]:
```
filtered_df = df.query('Age > 25')
filtered_df
```

[16]:
```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001        Aarohi  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..           …             …     …  …        …            …
194         3195         Meera  Female   34         HR      Rajasthan
195         3196        Aniket    Male   26    Finance    Uttarakhand
197         3198         Aryan    Male   36    Finance          Assam
198         3199       Karthik    Male   27    Finance          Assam
199         3200         Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                …            …                    …                …
194                24          78798                     10               Good
195                23         123696                      2          Excellent
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[171 rows x 10 columns]
```

[17]:
```
df_dropped = df.drop(columns=['Age'])
df_dropped
```

[17]:
```
     Employee_ID Employee_Name  Gender Department         Region  \
0           3001        Aarohi  Female      Sales          Assam
```

```
1          3002        Ananya  Female         IT          Bihar
2          3003        Ishita  Female    Finance  Uttar Pradesh
3          3004        Vihaan    Male  Marketing  Uttar Pradesh
4          3005         Dhruv    Male    Finance         Punjab
..          …           …       …          …            …
195        3196        Aniket    Male    Finance    Uttarakhand
196        3197         Dhruv    Male         HR         Odisha
197        3198         Aryan    Male    Finance          Assam
198        3199       Karthik    Male    Finance          Assam
199        3200         Sneha  Female         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                …              …                    …                …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 9 columns]
```

[18]:
```python
df_renamed = df.rename(columns={'Age':'Life Line'})
df_renamed
```

[18]:
```
     Employee_ID Employee_Name  Gender  Life Line Department         Region  \
0          3001        Aarohi  Female         36      Sales          Assam
1          3002        Ananya  Female         35         IT          Bihar
2          3003        Ishita  Female         45    Finance  Uttar Pradesh
3          3004        Vihaan    Male         56  Marketing  Uttar Pradesh
4          3005         Dhruv    Male         31    Finance         Punjab
..          …           …       …          …        …            …
195        3196        Aniket    Male         26    Finance    Uttarakhand
196        3197         Dhruv    Male         23         HR         Odisha
197        3198         Aryan    Male         36    Finance          Assam
198        3199       Karthik    Male         27    Finance          Assam
199        3200         Sneha  Female         34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
```

```
4                 4       46253                          7         Excellent
..              ...         ...                        ...               ...
195              23      123696                          2         Excellent
196              29      135434                          3              Good
197               2       75939                         10              Good
198               7       54516                          2         Excellent
199              11       81526                          3         Excellent

[200 rows x 10 columns]
```

[19]:
```python
df_sorted = df.sort_values(by='Age')
df_sorted
```

[19]:
```
     Employee_ID Employee_Name  Gender  Age  Department           Region  \
40          3041         Aarav    Male   21          HR        Telangana
153         3154         Aryan    Male   21       Sales            Assam
175         3176         Meera  Female   21          IT     Chhattisgarh
23          3024           Ira  Female   22  Operations        Jharkhand
111         3112        Ishaan    Male   22          IT      Uttarakhand
..           ...           ...     ...  ...         ...              ...
100         3101           Sai    Male   56          IT        Telangana
101         3102           Ira  Female   57          IT   Madhya Pradesh
76          3077         Meera  Female   57     Finance      Maharashtra
57          3058     Siddharth    Male   57          IT        Telangana
187         3188         Aditi  Female   57     Finance      West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
40                 11         109742                      8          Excellent
153                18          57602                      7          Excellent
175                21         100330                      2          Excellent
23                 24         100939                      3               Good
111                 8         112489                      6               Good
..                ...            ...                    ...                ...
100                25         102275                      3               Poor
101                 4         122756                      6               Good
76                  5         144796                      2               Poor
57                 12          42071                      9               Poor
187                 8          75100                      1            Average

[200 rows x 10 columns]
```

[20]:
```python
df_filled = df.fillna(0)
df_filled
```

[20]:
```
     Employee_ID Employee_Name  Gender  Age Department          Region  \
0           3001        Aarohi  Female   36      Sales           Assam
1           3002        Ananya  Female   35         IT           Bihar
```

```
2         3003        Ishita  Female   45    Finance  Uttar Pradesh
3         3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4         3005         Dhruv    Male   31    Finance         Punjab
..         …             …      …    …     …            …
195       3196        Aniket    Male   26    Finance    Uttarakhand
196       3197         Dhruv    Male   23         HR         Odisha
197       3198         Aryan    Male   36    Finance          Assam
198       3199       Karthik    Male   27    Finance          Assam
199       3200         Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                …             …                     …                …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[21]:
```python
df_unique = df.drop_duplicates()
df_unique
```

[21]:
```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001        Aarohi  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..         …             …      …    …     …            …
195         3196        Aniket    Male   26    Finance    Uttarakhand
196         3197         Dhruv    Male   23         HR         Odisha
197         3198         Aryan    Male   36    Finance          Assam
198         3199       Karthik    Male   27    Finance          Assam
199         3200         Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
```

```
..                 …          …                    …          …
195               23     123696                    2  Excellent
196               29     135434                    3       Good
197                2      75939                   10       Good
198                7      54516                    2  Excellent
199               11      81526                    3  Excellent

[200 rows x 10 columns]
```

[33]:
```
df_replaced = df.replace({'Aarohi':'Kavya'})
df_replaced
```

[33]:
```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001         Kavya  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..           …             …     …   …        …            …
195         3196       Aniket    Male   26    Finance    Uttarakhand
196         3197        Dhruv    Male   23         HR         Odisha
197         3198        Aryan    Male   36    Finance          Assam
198         3199      Karthik    Male   27    Finance          Assam
199         3200        Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                 …          …                    …          …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[23]:
```
grouped_df = df.groupby('Employee_Name').sum()
grouped_df
```

[23]:
```
               Employee_ID                                 Gender  \
Employee_Name
Aanya                18664     FemaleFemaleFemaleFemaleFemaleFemale
Aarav                21647            MaleMaleMaleMaleMaleMaleMale
```

```
Aarohi              18519          FemaleFemaleFemaleFemaleFemaleFemale
Abhishek             9350                                  MaleMaleMale
Aditi               24888  FemaleFemaleFemaleFemaleFemaleFemaleFemaleFemale
Aditya               9263                                  MaleMaleMale
Ananya              12290                      FemaleFemaleFemaleFemale
Aniket              18703                      MaleMaleMaleMaleMaleMale
Arjun                6334                                      MaleMale
Aryan               25016              MaleMaleMaleMaleMaleMaleMaleMale
Dhruv               21567                  MaleMaleMaleMaleMaleMaleMale
Diya                18740          FemaleFemaleFemaleFemaleFemaleFemale
Harsh                6132                                      MaleMale
Ira                 18383          FemaleFemaleFemaleFemaleFemaleFemale
Ishaan              12443                              MaleMaleMaleMale
Ishita               9133                            FemaleFemaleFemale
Karthik              9486                                  MaleMaleMale
Kavya                9260                            FemaleFemaleFemale
Krishna              6176                                      MaleMale
Manish               9439                                  MaleMaleMale
Meera               30866 FemaleFemaleFemaleFemaleFemaleFemaleFemaleFema…
Mira                18805          FemaleFemaleFemaleFemaleFemaleFemale
Neha                 6308                                  FemaleFemale
Nikhil               6111                                      MaleMale
Nisha               15422                FemaleFemaleFemaleFemaleFemale
Pooja               18527          FemaleFemaleFemaleFemaleFemaleFemale
Priya               24900  FemaleFemaleFemaleFemaleFemaleFemaleFemaleFemale
Rahul               18941                      MaleMaleMaleMaleMaleMale
Reyansh             27694          MaleMaleMaleMaleMaleMaleMaleMaleMale
Riya                18741          FemaleFemaleFemaleFemaleFemaleFemale
Rohan               12371                              MaleMaleMaleMale
Saanvi               9378                            FemaleFemaleFemale
Sai                 21656                  MaleMaleMaleMaleMaleMaleMale
Shreya              18391          FemaleFemaleFemaleFemaleFemaleFemale
Siddharth           15428                          MaleMaleMaleMaleMale
Simran              18630          FemaleFemaleFemaleFemaleFemaleFemale
Sneha               12438                      FemaleFemaleFemaleFemale
Tanvi               15499                FemaleFemaleFemaleFemaleFemale
Vihaan               6042                                      MaleMale
Vivaan              18519                      MaleMaleMaleMaleMaleMale

               Age                                 Department  \
Employee_Name
Aanya          232          MarketingFinanceMarketingITSalesMarketing
Aarav          247              HRHRHROperationsMarketingFinanceSales
Aarohi         227                     SalesFinanceITFinanceFinanceHR
Abhishek        93                                     OperationsITIT
Aditi          330              SalesITFinanceOperationsITITFinanceHR
Aditya         106                           OperationsSalesMarketing
```

```
Ananya          144                              ITSalesMarketingHR
Aniket          201                 MarketingHRHRITMarketingFinance
Arjun            70                                         SalesIT
Aryan           336     ITOperationsHRMarketingSalesFinanceITFinance
Dhruv           232            FinanceHROperationsMarketingHRSalesHR
Diya            233                     OperationsITSalesHRITSales
Harsh            89                                 SalesOperations
Ira             237               OperationsITFinanceMarketingITIT
Ishaan          149                         OperationsITMarketingIT
Ishita           95                     FinanceMarketingOperations
Karthik         121                             OperationsITFinance
Kavya            98                                      HRHRSales
Krishna          87                                           ITHR
Manish          120                     MarketingFinanceOperations
Meera           364  MarketingFinanceSalesITOperationsFinanceITSale…
Mira            293             MarketingFinanceSalesHRITOperations
Neha             92                                         HRSales
Nikhil           72                                         HRSales
Nisha           199                      MarketingOperationsITITSales
Pooja           285                      ITHRSalesFinanceFinanceSales
Priya           285           FinanceHRFinanceHRMarketingITSalesSales
Rahul           194                          ITITHRSalesITMarketing
Reyansh         403   OperationsHRHROperationsITITSalesFinanceFinance
Riya            246             SalesHRITMarketingMarketingFinance
Rohan           157                                 ITMarketingHRHR
Saanvi          110                                   MarketingHRIT
Sai             274     MarketingOperationsOperationsITMarketingITIT
Shreya          263                 OperationsITSalesHRMarketingHR
Siddharth       180                     MarketingITITMarketingSales
Simran          239           HRFinanceOperationsITMarketingMarketing
Sneha           143                             HRSalesOperationsHR
Tanvi           155                       HRHRITOperationsFinance
Vihaan           96                                  MarketingSales
Vivaan          222     MarketingSalesFinanceOperationsOperationsHR

                                                          Region  \
Employee_Name
Aanya                TelanganaDelhiOdishaAssamGujaratUttarakhand
Aarav          GujaratTelanganaOdishaBiharRajasthanMadhya Pra…
Aarohi           AssamKeralaDelhiWest BengalMaharashtraRajasthan
Abhishek               Uttar PradeshUttar PradeshUttarakhand
Aditi          ChhattisgarhUttarakhandPunjabKarnatakaHaryanaO…
Aditya                               HaryanaJharkhandOdisha
Ananya                         BiharKarnatakaAssamMadhya Pradesh
Aniket           BiharPunjabKarnatakaGujaratJharkhandUttarakhand
Arjun                                Uttar PradeshTamil Nadu
Aryan          KarnatakaKeralaChhattisgarhUttarakhandAssamBih…
```

Dhruv	PunjabAssamHaryanaMadhya PradeshMaharashtraDel…
Diya	HaryanaUttar PradeshTamil NaduGujaratTelangana…
Harsh	TelanganaKerala
Ira	JharkhandWest BengalKeralaAndhra PradeshMadhya…
Ishaan	West BengalUttarakhandTelanganaBihar
Ishita	Uttar PradeshUttarakhandGujarat
Karthik	OdishaAssamAssam
Kavya	GujaratPunjabKerala
Krishna	JharkhandChhattisgarh
Manish	Andhra PradeshTamil NaduUttarakhand
Meera	West BengalRajasthanAndhra PradeshAndhra Prade…
Mira	TelanganaAndhra PradeshRajasthanAndhra Pradesh…
Neha	DelhiGujarat
Nikhil	DelhiAndhra Pradesh
Nisha	PunjabOdishaTamil NaduTelanganaTelangana
Pooja	BiharPunjabAssamTamil NaduDelhiGujarat
Priya	PunjabHaryanaPunjabTamil NaduAssamUttarakhandM…
Rahul	GujaratGujaratRajasthanAndhra PradeshUttar Pra…
Reyansh	ChhattisgarhUttarakhandJharkhandMadhya Pradesh…
Riya	HaryanaPunjabWest BengalGujaratAssamAndhra Pra…
Rohan	PunjabJharkhandWest BengalDelhi
Saanvi	West BengalDelhiAndhra Pradesh
Sai	PunjabGujaratKeralaTelanganaTelanganaChhattisg…
Shreya	Tamil NaduUttarakhandKeralaMadhya PradeshUttar…
Siddharth	KarnatakaTelanganaWest BengalTelanganaRajasthan
Simran	JharkhandOdishaBiharKarnatakaMaharashtraJharkhand
Sneha	Tamil NaduAndhra PradeshHaryanaWest Bengal
Tanvi	KeralaAndhra PradeshPunjabWest BengalUttar Pra…
Vihaan	Uttar PradeshJharkhand
Vivaan	Andhra PradeshUttarakhandOdishaChhattisgarhKar…

	Experience_Years	Monthly_Sales	Customer_Satisfaction	\
Employee_Name				
Aanya	90	435857	40	
Aarav	73	541468	30	
Aarohi	74	358947	33	
Abhishek	21	195094	7	
Aditi	120	612759	47	
Aditya	67	206280	18	
Ananya	33	247081	25	
Aniket	51	523055	34	
Arjun	29	142425	12	
Aryan	130	532638	55	
Dhruv	90	529652	28	
Diya	77	535873	29	
Harsh	30	131797	13	
Ira	83	625345	32	

Ishaan	56	358236	27
Ishita	53	150761	14
Karthik	21	205527	13
Kavya	27	269336	22
Krishna	17	53640	17
Manish	68	261012	14
Meera	180	993714	50
Mira	79	683913	34
Neha	26	166518	16
Nikhil	54	119517	14
Nisha	75	383939	20
Pooja	97	497282	22
Priya	155	859827	45
Rahul	122	484471	27
Reyansh	150	768804	45
Riya	48	547009	21
Rohan	56	294265	19
Saanvi	44	320296	21
Sai	87	654477	46
Shreya	133	501446	31
Siddharth	44	285302	36
Simran	111	430843	22
Sneha	81	162589	23
Tanvi	98	492757	34
Vihaan	38	154103	12
Vivaan	96	569272	26

	Performance_Rating
Employee_Name	
Aanya	ExcellentGoodGoodPoorAverageGood
Aarav	ExcellentExcellentExcellentExcellentExcellentA…
Aarohi	AverageAverageAverageExcellentExcellentPoor
Abhishek	ExcellentAverageGood
Aditi	AverageExcellentGoodExcellentGoodAverageAverag…
Aditya	GoodExcellentGood
Ananya	GoodAverageGoodExcellent
Aniket	ExcellentGoodExcellentGoodGoodExcellent
Arjun	GoodGood
Aryan	AverageExcellentExcellentGoodExcellentGoodAver…
Dhruv	ExcellentExcellentExcellentExcellentExcellentG…
Diya	GoodExcellentExcellentExcellentAverageAverage
Harsh	AverageAverage
Ira	GoodExcellentGoodGoodGoodExcellent
Ishaan	ExcellentGoodGoodExcellent
Ishita	AverageGoodGood
Karthik	AverageAverageExcellent
Kavya	GoodGoodAverage

```
Krishna                                                  ExcellentExcellent
Manish                                                  GoodGoodExcellent
Meera           AverageExcellentExcellentGoodGoodPoorGoodGoodE…
Mira                 GoodExcellentExcellentGoodExcellentExcellent
Neha                                                             GoodGood
Nikhil                                                   ExcellentExcellent
Nisha                                    ExcellentAverageAveragePoorPoor
Pooja                                GoodGoodAverageExcellentGoodAverage
Priya           GoodGoodExcellentExcellentAverageGoodExcellent…
Rahul                                        GoodAverageGoodGoodGoodGood
Reyansh         ExcellentExcellentGoodExcellentExcellentAverag…
Riya                                 GoodAverageExcellentGoodAveragePoor
Rohan                                                GoodGoodAverageGood
Saanvi                                       ExcellentExcellentExcellent
Sai                                   GoodGoodGoodPoorAverageGoodAverage
Shreya                                       GoodGoodGoodGoodGoodAverage
Siddharth                                           GoodPoorGoodGoodGood
Simran                               GoodGoodAverageExcellentGoodAverage
Sneha                                              GoodPoorGoodExcellent
Tanvi                                    AveragePoorExcellentPoorAverage
Vihaan                                                           GoodGood
Vivaan                           GoodExcellentPoorExcellentPoorExcellent
```

[24]:
```python
agg_df = df.groupby('Employee_Name').agg({'Age':'mean'})
agg_df
```

[24]:
```
                     Age
Employee_Name
Aanya          38.666667
Aarav          35.285714
Aarohi         37.833333
Abhishek       31.000000
Aditi          41.250000
Aditya         35.333333
Ananya         36.000000
Aniket         33.500000
Arjun          35.000000
Aryan          42.000000
Dhruv          33.142857
Diya           38.833333
Harsh          44.500000
Ira            39.500000
Ishaan         37.250000
Ishita         31.666667
Karthik        40.333333
Kavya          32.666667
Krishna        43.500000
```

```
Manish         40.000000
Meera          36.400000
Mira           48.833333
Neha           46.000000
Nikhil         36.000000
Nisha          39.800000
Pooja          47.500000
Priya          35.625000
Rahul          32.333333
Reyansh        44.777778
Riya           41.000000
Rohan          39.250000
Saanvi         36.666667
Sai            39.142857
Shreya         43.833333
Siddharth      36.000000
Simran         39.833333
Sneha          35.750000
Tanvi          31.000000
Vihaan         48.000000
Vivaan         37.000000
```

[25]:
```python
agg_df = df.groupby('Employee_Name').agg({'Monthly_Sales':'mean'})
agg_df
```

[25]:
```
               Monthly_Sales
Employee_Name
Aanya           72642.833333
Aarav           77352.571429
Aarohi          59824.500000
Abhishek        65031.333333
Aditi           76594.875000
Aditya          68760.000000
Ananya          61770.250000
Aniket          87175.833333
Arjun           71212.500000
Aryan           66579.750000
Dhruv           75664.571429
Diya            89312.166667
Harsh           65898.500000
Ira            104224.166667
Ishaan          89559.000000
Ishita          50253.666667
Karthik         68509.000000
Kavya           89778.666667
Krishna         26820.000000
Manish          87004.000000
```

```
Meera           99371.400000
Mira           113985.500000
Neha            83259.000000
Nikhil          59758.500000
Nisha           76787.800000
Pooja           82880.333333
Priya          107478.375000
Rahul           80745.166667
Reyansh         85422.666667
Riya            91168.166667
Rohan           73566.250000
Saanvi         106765.333333
Sai             93496.714286
Shreya          83574.333333
Siddharth       57060.400000
Simran          71807.166667
Sneha           40647.250000
Tanvi           98551.400000
Vihaan          77051.500000
Vivaan          94878.666667
```

[26]:
```python
count_df = df.groupby('Monthly_Sales').count()
count_df
```

[26]:
```
               Employee_ID  Employee_Name  Gender  Age  Department  Region  \
Monthly_Sales
25155                    1              1       1    1           1       1
25282                    1              1       1    1           1       1
25587                    1              1       1    1           1       1
25645                    1              1       1    1           1       1
26089                    1              1       1    1           1       1
…                      …            …      …   …         …      …
147046                   1              1       1    1           1       1
147232                   1              1       1    1           1       1
147775                   1              1       1    1           1       1
148950                   1              1       1    1           1       1
149661                   1              1       1    1           1       1

               Experience_Years  Customer_Satisfaction  Performance_Rating
Monthly_Sales
25155                         1                      1                   1
25282                         1                      1                   1
25587                         1                      1                   1
25645                         1                      1                   1
26089                         1                      1                   1
…                          …                    …                  …
147046                        1                      1                   1
```

```
147232                      1                      1                    1
147775                      1                      1                    1
148950                      1                      1                    1
149661                      1                      1                    1

[200 rows x 9 columns]
```

[27]:

```
sum_df = df.groupby('Age').sum()
sum_df
```

[27]:

```
     Employee_ID                                     Employee_Name  \
Age
21          9371                                  AaravAryanMeera
22         21817                    IraDhruvIshaanRahulRahulAarohiMeera
23         21960                  IshitaVivaanAnanyaAaravSaiSaanviDhruv
24         24534           MeeraSiddharthAaravPriyaAditiAniketTanviDiya
25         12469                         AdityaSiddharthRahulAbhishek
26         15768                           AniketAbhishekAanyaPriyaAniket
27         18478                   AditiIshitaRohanTanviShreyaKarthik
28         24851                  KavyaDhruvSnehaIshaanMeeraRiyaSaiAditi
29         15462                             AanyaSnehaKavyaAanyaVivaan
30         15299                           PoojaAditiAniketNikhilTanvi
31         15390                               DhruvNishaDiyaTanviRiya
32         12639                                  DiyaArjunSimranPriya
33         21443                  ReyanshIraSaanviReyanshAryanHarshNisha
34         15661                         RiyaSiddharthPriyaMeeraSneha
35         15322                         AnanyaMeeraReyanshPriyaRahul
36         28007     AarohiSaiSimranRohanAnanyaManishVivaanSimranAryan
37          9343                                          SimranIraSai
38         18713                    AdityaVivaanShreyaKrishnaMiraArjun
39         12235                                  SaiDhruvRohanAarohi
40         12494                           VihaanNehaNishaSiddharth
41         21662                  VivaanAarohiAaravNishaDhruvKavyaManish
42         27958     ReyanshNikhilAryanAbhishekAanyaAaravPriyaDiyaR…
43         15752                       ManishTanviKarthikAdityaAarohi
44         15455                                 IraIraMiraPoojaRahul
45         12183                               IshitaPriyaMeeraAniket
46         15410                         AarohiShreyaSimranAaravRahul
47         12360                               MeeraPriyaIshaanPooja
48         12294                               ShreyaDhruvShreyaRiya
49          6152                                         KrishnaRiya
50         25010                DiyaAnanyaMiraAryanAniketAaravAryanMira
51          9204                                  AryanMeeraKarthik
52         12459                               SnehaIshaanSimranNeha
53         18685                      AanyaReyanshAditiPoojaAanyaAryan
54         12378                              NishaReyanshDiyaSaanvi
55         21760                   VivaanMiraPoojaSaiRohanReyanshAditi
```

```
56        27697      VihaanHarshPoojaShreyaReyanshSaiAditiRiyaMira
57        12425                             SiddharthMeeraIraAditi

                                            Gender  \
Age
21                                     MaleMaleFemale
22                 FemaleMaleMaleMaleMaleFemaleFemale
23                 FemaleMaleFemaleMaleMaleFemaleMale
24         FemaleMaleMaleFemaleFemaleMaleFemaleFemale
25                                       MaleMaleMaleMale
26                              MaleMaleFemaleFemaleMale
27                   FemaleFemaleMaleFemaleFemaleMale
28         FemaleMaleFemaleMaleFemaleFemaleMaleFemale
29                       FemaleFemaleFemaleFemaleMale
30                         FemaleFemaleMaleMaleFemale
31                       MaleFemaleFemaleFemaleFemale
32                             FemaleMaleFemaleFemale
33                 MaleFemaleFemaleMaleMaleMaleFemale
34                       FemaleMaleFemaleFemaleFemale
35                         FemaleFemaleMaleFemaleMale
36       FemaleMaleFemaleMaleFemaleMaleMaleFemaleMale
37                                       FemaleFemaleMale
38                       MaleMaleFemaleMaleFemaleMale
39                                     MaleMaleMaleFemale
40                                   MaleFemaleFemaleMale
41                 MaleFemaleMaleFemaleMaleFemaleMale
42           MaleMaleMaleMaleFemaleMaleFemaleFemaleMale
43                             MaleFemaleMaleMaleFemale
44                       FemaleFemaleFemaleFemaleMale
45                             FemaleFemaleFemaleMale
46                         FemaleFemaleFemaleMaleMale
47                             FemaleFemaleMaleFemale
48                             FemaleMaleFemaleFemale
49                                         MaleFemale
50           FemaleFemaleFemaleMaleMaleMaleMaleFemale
51                                     MaleFemaleMale
52                             FemaleMaleFemaleFemale
53                   FemaleMaleFemaleFemaleFemaleMale
54                             FemaleMaleFemaleFemale
55                 MaleFemaleFemaleMaleMaleMaleFemale
56     MaleMaleFemaleFemaleMaleMaleFemaleFemaleFemale
57                             MaleFemaleFemaleFemale

                                            Department  \
Age
21                                           HRSalesIT
22                         OperationsHRITITITFinanceSales
```

```
23        OperationsOperationsMarketingMarketingITITHR
24           FinanceMarketingHRHRFinanceHROperationsIT
25                                 SalesMarketingHRIT
26                              ITITSalesSalesFinance
27                  SalesMarketingITHRMarketingFinance
28                    HROperationsOperationsITITHRITIT
29                            MarketingHRHRMarketingHR
30                                ITITMarketingSalesHR
31                        FinanceOperationsITITFinance
32                              HRSalesMarketingSales
33               OperationsITMarketingHRHROperationsIT
34                                      SalesITITHRHR
35                           ITMarketingOperationsHRIT
36   SalesOperationsHRHRHRFinanceOperationsMarketin…
37                              OperationsITMarketing
38                           OperationsFinanceHRHRITIT
39                  MarketingMarketingMarketingFinance
40                                    SalesHRSalesSales
41                 MarketingITHRITSalesSalesOperations
42   HRHROperationsOperationsITFinanceMarketingSale…
43                      MarketingFinanceITMarketingHR
44           FinanceMarketingMarketingFinanceMarketing
45                         FinanceFinanceOperationsHR
46                    FinanceITFinanceOperationsSales
47                         SalesFinanceMarketingSales
48                            OperationsHRHRMarketing
49                                         ITMarketing
50   OperationsSalesSalesMarketingMarketingSalesITO…
51                                     ITITOperations
52                              SalesOperationsITSales
53       FinanceSalesOperationsFinanceMarketingFinance
54                                 MarketingITSalesHR
55                SalesFinanceSalesOperationsHRFinanceHR
56                    MarketingSalesHRSalesITITITITHR
57                                  ITFinanceITFinance

                                                Region  Experience_Years  \
Age
21                        TelanganaAssamChhattisgarh                50
22   JharkhandMaharashtraUttarakhandGujaratGujaratM…               123
23   GujaratChhattisgarhAssamRajasthanGujaratAndhra…               116
24   RajasthanKarnatakaGujaratHaryanaPunjabKarnatak…                90
25              JharkhandTelanganaRajasthanUttarakhand              55
26   GujaratUttar PradeshGujaratMadhya PradeshUttar…                64
27   ChhattisgarhUttarakhandPunjabKeralaUttar Prade…                98
28   GujaratHaryanaHaryanaBiharChhattisgarhPunjabCh…                94
29                 TelanganaTamil NaduPunjabOdishaDelhi             81
```

```
30   BiharUttarakhandBiharAndhra PradeshAndhra Pradesh                  89
31       PunjabOdishaUttar PradeshPunjabAndhra Pradesh                  93
32              GujaratUttar PradeshJharkhandJharkhand                  20
33   ChhattisgarhWest BengalWest BengalJharkhandChh…                  111
34   HaryanaWest BengalUttarakhandRajasthanWest Bengal                  62
35   BiharWest BengalMadhya PradeshTamil NaduUttar …                   94
36   AssamGujaratJharkhandDelhiMadhya PradeshTamil …                  126
37                             BiharDelhiTelangana                      63
38   HaryanaOdishaMaharashtraChhattisgarhBiharTamil…                  117
39            PunjabMadhya PradeshJharkhandWest Bengal                  53
40                    JharkhandDelhiTelanganaRajasthan                  77
41   Andhra PradeshDelhiOdishaTamil NaduDelhiKerala…                   87
42   UttarakhandDelhiKeralaUttar PradeshAssamMadhya…                  174
43     Andhra PradeshUttar PradeshAssamOdishaRajasthan                  84
44   KeralaAndhra PradeshTelanganaTamil NaduUttarak…                   71
45                      Uttar PradeshPunjabBiharPunjab                  78
46         KeralaUttarakhandOdishaBiharAndhra Pradesh                   61
47               Andhra PradeshPunjabTelanganaGujarat                   79
48                 Tamil NaduAssamMadhya PradeshAssam                   59
49                                    JharkhandGujarat                   4
50   HaryanaKarnatakaRajasthanUttarakhandJharkhandO…                   83
51                       KarnatakaAndhra PradeshOdisha                  55
52          Andhra PradeshWest BengalKarnatakaGujarat                   66
53          DelhiKeralaKarnatakaDelhiUttarakhandBihar                  103
54                          PunjabAssamTamil NaduDelhi                  36
55   UttarakhandAndhra PradeshAssamKeralaWest Benga…                  109
56   Uttar PradeshTelanganaPunjabKeralaKarnatakaTel…                  130
57       TelanganaMaharashtraMadhya PradeshWest Bengal                  29

     Monthly_Sales  Customer_Satisfaction  \
Age
21          267674                     17
22          566720                     45
23          694950                     32
24          625812                     46
25          368506                     20
26          455680                     19
27          420960                     27
28          699579                     57
29          462089                     26
30          347780                     32
31          444305                     22
32          489034                     21
33          494887                     31
34          370381                     33
35          308808                     26
36          700359                     44
```

37	210333	21
38	527407	31
39	276991	21
40	326085	24
41	411529	34
42	750453	44
43	442586	22
44	564361	25
45	239111	15
46	263260	29
47	461286	17
48	309339	28
49	161640	10
50	585257	53
51	301110	22
52	217457	24
53	397150	28
54	497818	26
55	657645	34
56	584062	50
57	384723	18

	Performance_Rating
Age	
21	ExcellentExcellentExcellent
22	GoodExcellentGoodGoodAverageExcellentGood
23	GoodExcellentGoodExcellentAverageExcellentGood
24	ExcellentGoodExcellentGoodGoodExcellentPoorAve…
25	ExcellentGoodGoodGood
26	GoodAverageAverageExcellentExcellent
27	AverageGoodGoodAverageGoodExcellent
28	GoodExcellentGoodExcellentGoodAverageGoodAverage
29	ExcellentGoodGoodGoodExcellent
30	GoodExcellentExcellentExcellentPoor
31	ExcellentAverageExcellentExcellentPoor
32	ExcellentGoodAverageAverage
33	ExcellentExcellentExcellentGoodExcellentAverag…
34	GoodGoodGoodGoodExcellent
35	GoodAverageExcellentExcellentGood
36	AverageGoodGoodGoodExcellentGoodPoorGoodGood
37	AverageExcellentAverage
38	GoodPoorAverageExcellentExcellentGood
39	GoodExcellentGoodExcellent
40	GoodGoodPoorGood
41	GoodAverageExcellentAverageGoodAverageExcellent
42	ExcellentExcellentExcellentExcellentPoorAverag…
43	GoodAverageAverageGoodPoor

```
44                       GoodGoodGoodExcellentGood
45                          AverageGoodGoodGood
46                  AverageGoodGoodExcellentGood
47                 ExcellentExcellentGoodAverage
48                      GoodExcellentGoodAverage
49                                 ExcellentGood
50    GoodAverageExcellentGoodGoodExcellentAverageEx…
51                           AverageGoodAverage
52                    PoorExcellentExcellentGood
53                 GoodGoodExcellentGoodGoodGood
54            ExcellentAverageExcellentExcellent
55     ExcellentExcellentAverageGoodAverageGoodAverage
56    GoodAverageGoodGoodExcellentPoorGoodExcellentGood
57                          PoorPoorGoodAverage
```

[28]:
```python
max_df = df.groupby('Monthly_Sales').max()
max_df
```

[28]:
```
               Employee_ID Employee_Name  Gender  Age  Department  \
Monthly_Sales
25155                 3164        Simran  Female   36   Marketing
25282                 3046         Dhruv    Male   48          HR
25587                 3122      Abhishek    Male   26          IT
25645                 3107         Sneha  Female   28  Operations
26089                 3160       Krishna    Male   38          HR
…                        …           …       …    …         …
147046                3183         Priya  Female   26       Sales
147232                3110         Tanvi  Female   24  Operations
147775                3020           Sai    Male   36  Operations
148950                3165        Simran  Female   32   Marketing
149661                3065        Vivaan    Male   38     Finance

                       Region  Experience_Years  Customer_Satisfaction  \
Monthly_Sales
25155             Maharashtra                26                      5
25282                   Assam                 6                      4
25587           Uttar Pradesh                 0                      1
25645                 Haryana                24                      9
26089            Chhattisgarh                14                      8
…                         …                …                      …
147046         Madhya Pradesh                19                      5
147232            West Bengal                24                      7
147775                Gujarat                17                      6
148950              Jharkhand                 9                      1
149661                 Odisha                10                      5

              Performance_Rating
```

```
Monthly_Sales
25155                          Good
25282                     Excellent
25587                       Average
25645                          Good
26089                     Excellent
…                                 …
147046                    Excellent
147232                         Poor
147775                         Good
148950                      Average
149661                         Poor

[200 rows x 9 columns]
```

[29]:

```
min_df = df.groupby('Monthly_Sales').min()
min_df
```

[29]:

```
               Employee_ID Employee_Name  Gender  Age  Department  \
Monthly_Sales
25155                 3164        Simran  Female   36   Marketing
25282                 3046         Dhruv    Male   48          HR
25587                 3122      Abhishek    Male   26          IT
25645                 3107         Sneha  Female   28  Operations
26089                 3160       Krishna    Male   38          HR
…                        …           …       …  …           …
147046                3183         Priya  Female   26       Sales
147232                3110         Tanvi  Female   24  Operations
147775                3020           Sai    Male   36  Operations
148950                3165        Simran  Female   32   Marketing
149661                3065        Vivaan    Male   38     Finance

                       Region  Experience_Years  Customer_Satisfaction  \
Monthly_Sales
25155             Maharashtra                26                      5
25282                   Assam                 6                      4
25587           Uttar Pradesh                 0                      1
25645                 Haryana                24                      9
26089            Chhattisgarh                14                      8
…                           …                 …                     …
147046         Madhya Pradesh                19                      5
147232            West Bengal                24                      7
147775                Gujarat                17                      6
148950              Jharkhand                 9                      1
149661                 Odisha                10                      5

              Performance_Rating
```

```
Monthly_Sales
25155                   Good
25282              Excellent
25587                Average
25645                   Good
26089              Excellent
…                          …
147046             Excellent
147232                  Poor
147775                  Good
148950               Average
149661                  Poor

[200 rows x 9 columns]
```

[30]:

```python
df_cleaned = df.dropna()
df_cleaned
```

[30]:

```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001        Aarohi  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..           …             …     …    …   …          …
195         3196        Aniket    Male   26    Finance    Uttarakhand
196         3197         Dhruv    Male   23         HR         Odisha
197         3198         Aryan    Male   36    Finance          Assam
198         3199       Karthik    Male   27    Finance          Assam
199         3200         Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                …            …                    …                …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[31]:
```
df_filled = df.fillna(0)
df_filled
```

[31]:
```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001        Aarohi  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..           …           …      …   …        …            …
195         3196        Aniket    Male   26    Finance    Uttarakhand
196         3197         Dhruv    Male   23         HR         Odisha
197         3198         Aryan    Male   36    Finance          Assam
198         3199       Karthik    Male   27    Finance          Assam
199         3200         Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                …            …                      …                …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[32]:
```
df_replaced = df.replace({'Aarohi':100})
df_replaced
```

[32]:
```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001           100  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..           …           …      …   …        …            …
195         3196        Aniket    Male   26    Finance    Uttarakhand
196         3197         Dhruv    Male   23         HR         Odisha
197         3198         Aryan    Male   36    Finance          Assam
198         3199       Karthik    Male   27    Finance          Assam
199         3200         Sneha  Female   34         HR    West Bengal
```

```
     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                  …              …                      …                  …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[35]:

```
pip install matplotlib
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: matplotlib in
c:\users\hp\appdata\roaming\python\python313\site-packages (3.10.7)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(1.4.9)
Requirement already satisfied: numpy>=1.23 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(2.3.4)
Requirement already satisfied: packaging>=20.0 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(25.0)
Requirement already satisfied: pillow>=8 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(12.0.0)
Requirement already satisfied: pyparsing>=3 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib)
(2.9.0.post0)
```

```
Requirement already satisfied: six>=1.5 in
c:\users\hp\appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.7->matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

[36]:
```
df
```

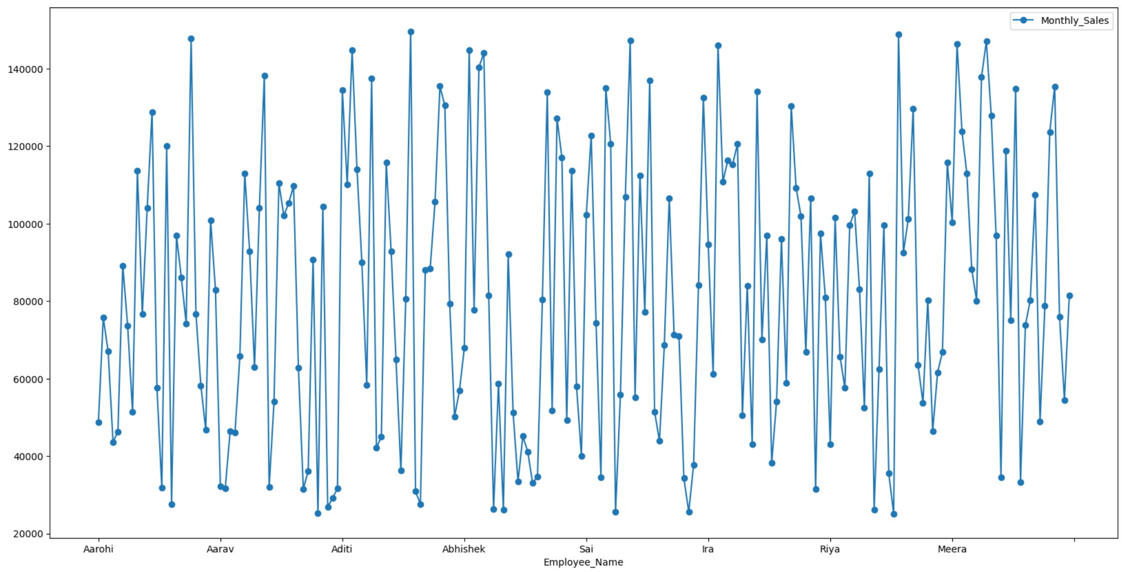
[36]:
```
     Employee_ID Employee_Name  Gender  Age Department         Region  \
0           3001        Aarohi  Female   36      Sales          Assam
1           3002        Ananya  Female   35         IT          Bihar
2           3003        Ishita  Female   45    Finance  Uttar Pradesh
3           3004        Vihaan    Male   56  Marketing  Uttar Pradesh
4           3005         Dhruv    Male   31    Finance         Punjab
..           …            …     …    …        …             …
195         3196        Aniket    Male   26    Finance    Uttarakhand
196         3197         Dhruv    Male   23         HR         Odisha
197         3198         Aryan    Male   36    Finance          Assam
198         3199       Karthik    Male   27    Finance          Assam
199         3200         Sneha  Female   34         HR    West Bengal

     Experience_Years  Monthly_Sales  Customer_Satisfaction Performance_Rating
0                   2          48648                      2            Average
1                  13          75729                      3               Good
2                  25          66971                      1            Average
3                  18          43658                      6               Good
4                   4          46253                      7          Excellent
..                …            …                    …                …
195                23         123696                      2          Excellent
196                29         135434                      3               Good
197                 2          75939                     10               Good
198                 7          54516                      2          Excellent
199                11          81526                      3          Excellent

[200 rows x 10 columns]
```

[ ]:
```
import matplotlib.pyplot as plt
```

```
Matplotlib is building the font cache; this may take a moment.
```

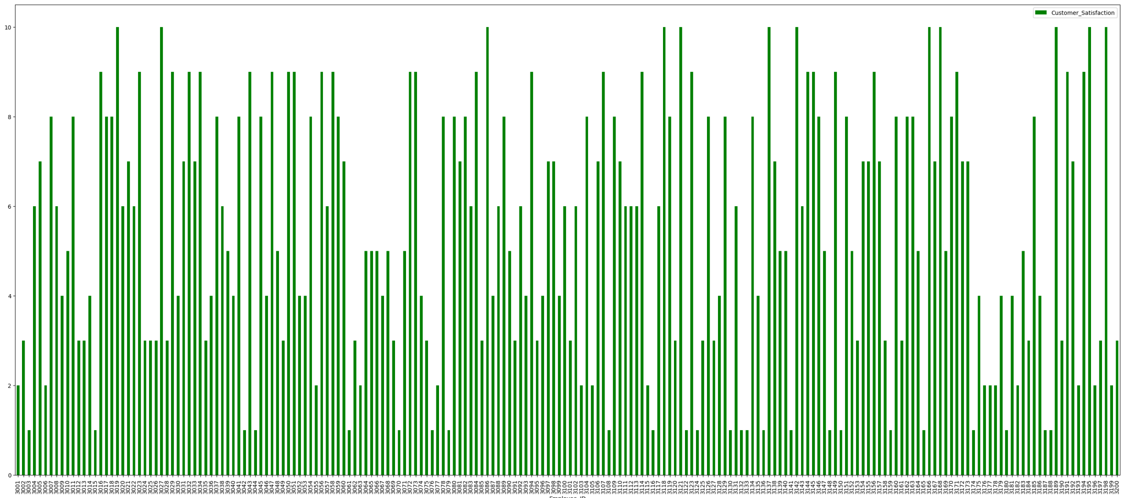
[39]:
```
df.
↪plot(x='Employee_Name',y='Monthly_Sales',kind='line',marker='o',figsize=(20,10))
```

[39]:
```
<Axes: xlabel='Employee_Name'>
```



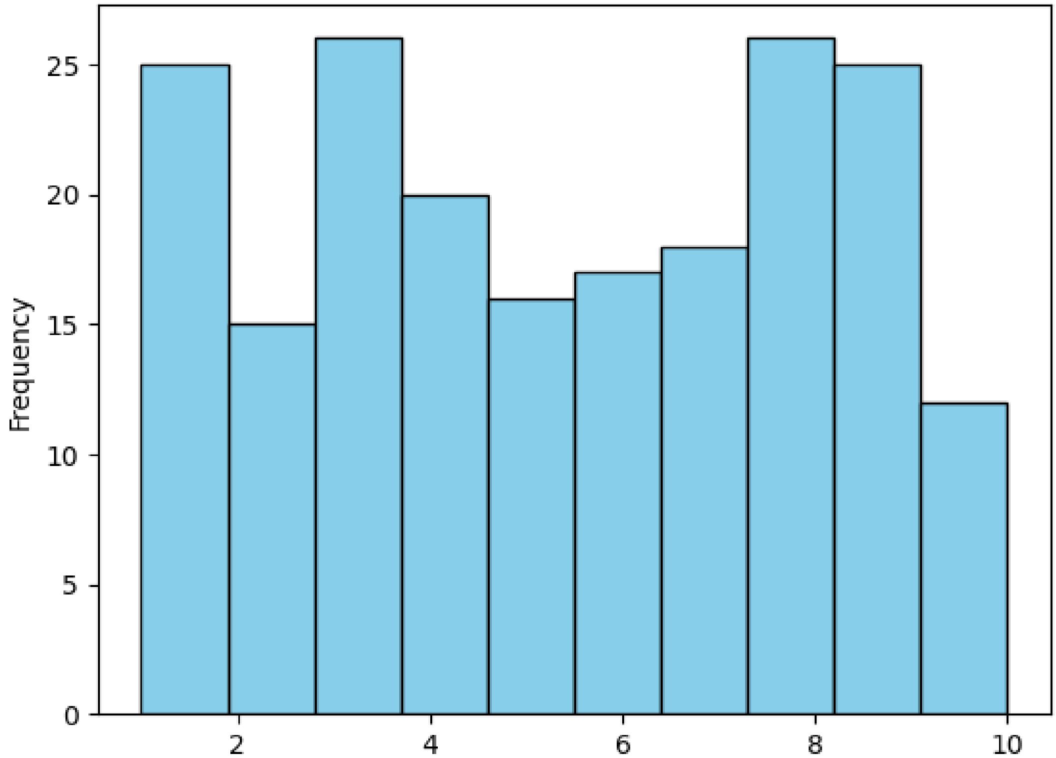
[66]:
```
df.plot(x='Employee_ID', y='Customer_Satisfaction', kind='bar', color='green',␣
↪figsize=(35, 15))
```

[66]: <Axes: xlabel='Employee_ID'>



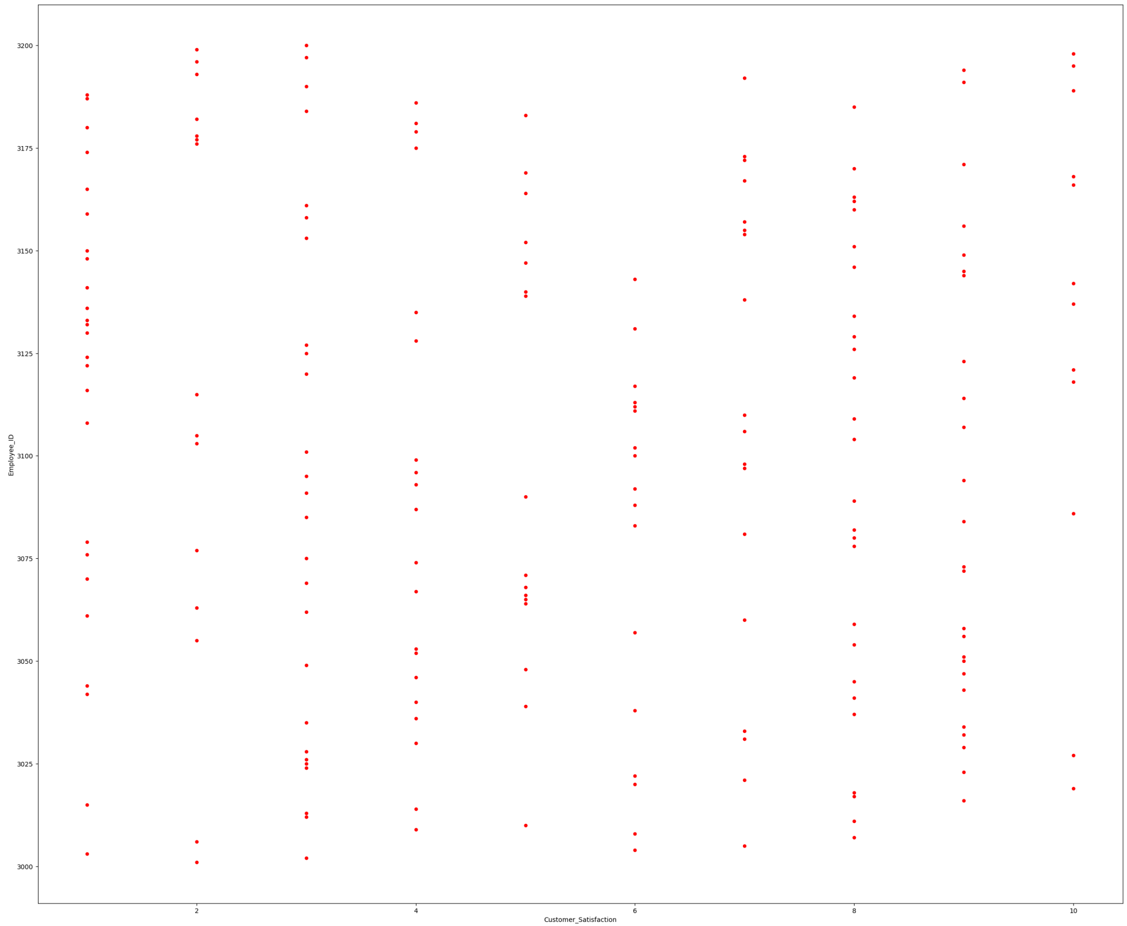
[53]:
```
df['Customer_Satisfaction'].plot(kind='hist',color='skyblue', edgecolor='black')
```

[53]: <Axes: ylabel='Frequency'>




```
[67]: df.plot(x='Customer_Satisfaction', y='Employee_ID',kind='scatter',color='red',␣
      ↪figsize=(30,25))
```

```
[67]: <Axes: xlabel='Customer_Satisfaction', ylabel='Employee_ID'>
```



3200
3175
3150
3125
3100
3075
3050
3025
3000
Employee_ID
2
4
6
8
10
Customer_Satisfaction